

From weighted path graphs to Sturm-Liouville problems

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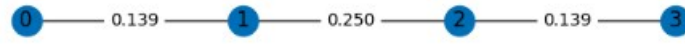
Abstract. The goal of this note is to present a continuum approach to an eigenvalue problem for weighted path graphs.

Problem. Let us consider the (undirected) path graph with vertices $V = \{0, 1, \dots, n-1\}$. Define the edge weights by

$$w_{k,k+1} = \frac{(k + \frac{1}{2})(n - k - \frac{3}{2})}{(n-1)^2}, \quad k = 0, 1, \dots, n-2.$$

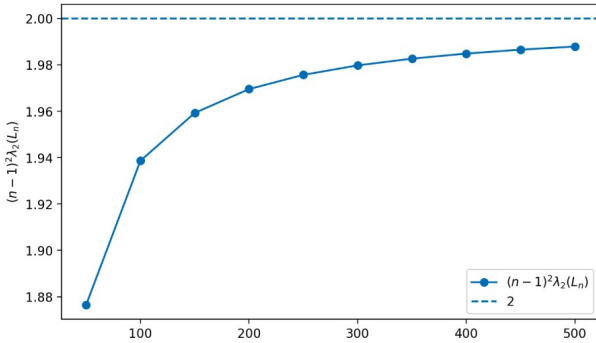
Let L_n be the combinatorial Laplacian of the weighted graph. We show that the principal eigenvalue (spectral gap) satisfies

$$\lambda_2(L_n) \sim \frac{2}{(n-1)^2} \quad \text{as } n \rightarrow \infty.$$

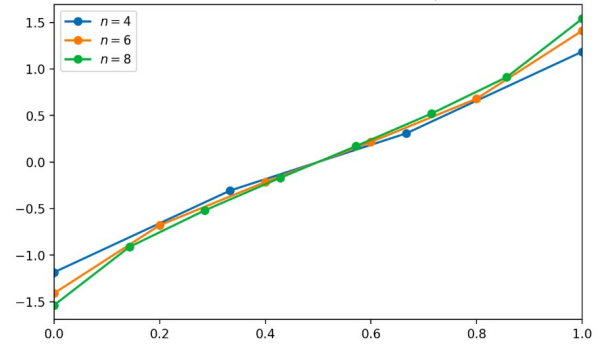


Weighted path graph with $n = 4$

Before providing a solution to the problem, recall that the combinatorial Laplacian is defined by $L_n = D_n - A_n$, where D_n is a diagonal matrix with the sum of the weights at each vertex, respectively, and A_n is the adjacency matrix of the weighted path graph.



Values $(n-1)^2 \lambda_2(L_n)$ for $n = 50, 100, \dots, 500$



Corresponding unit-length eigenvectors for $n = 4, 6$ and 8 , rescaled by $\sqrt{n-1}$

Solution. According to the Rayleigh-Ritz formula for matrix eigenvalues, the principal eigenvalue of the Laplacian is given by

$$\lambda_2(L_n) = \min_{u \perp 1, u \neq 0} \frac{u^T L_n u}{u^T u}.$$

To find the limit of these spectral gaps, we study the corresponding continuum variational problem. Note that

$$u^T L_n u = \sum_{k=0}^{n-2} w_{k,k+1} (u_k - u_{k+1})^2.$$

Since $w_{k,k+1}$ -s are the midpoint values of the function $w(x) = x(1-x)$ on the intervals $(k/(n-1), (k+1)/(n-1))$ and $(u_k - u_{k+1})_k$ is a discrete differential, we naturally introduce the weighted Sobolev space

$$H_w^1(0, 1) = \left\{ f \in L^2(0, 1) : \int_0^1 x(1-x) |f'(x)|^2 dx < \infty \right\},$$

the functional

$$\mathcal{E}(f) = \frac{\int_0^1 w(x) f'(x)^2 dx}{\int_0^1 f^2(x) dx},$$

and the variational problem

$$\inf_{f \in \mathcal{A}} \frac{\int_0^1 w(x) f'(x)^2 dx}{\int_0^1 f^2(x) dx} = \inf_{f \in \mathcal{A}} \mathcal{E}(f),$$

where

$$\mathcal{A} = \left\{ f \in H_w^1(0, 1) : \int_0^1 f(x) dx = 0 \text{ and } f \not\equiv 0 \right\}.$$

To justify the passage from the discrete setting to the continuum limit, we divide the argument into several steps. After identifying the limiting variational problem, we solve it via the corresponding Euler-Lagrange equation, which leads to the classical Legendre problem and its smallest nonzero eigenvalue.

We begin by collecting a few basic facts about the weighted Sobolev space $H_w^1(0, 1)$. It is a Hilbert space with the inner product

$$(f, g) = \int_0^1 f g dx + \int_0^1 f' g' w dx.$$

Moreover, the Legendre polynomials $L_n(x/2 + 1/2)$ form an orthogonal basis of $H_w^1(0, 1)$. In addition, a Rellich-Kondrachov-type compactness theorem is still valid in $H_w^1(0, 1)$. That is, if $\{f_n\}_n$ is a bounded sequence in $H_w^1(0, 1)$, then it has a subsequence converging to a function $f \in L^2(0, 1)$ in $L^2(0, 1)$. For the proofs of these statements, we refer the reader to the Appendix.

We continue with the upper bound, that is, the lim sup inequality for the rescaled spectral gaps.

Step 1. We prove

$$\limsup_{n \rightarrow \infty} \left((n-1)^2 \min_{u \perp 1, u \neq 0} \frac{u^T L_n u}{u^T u} \right) \leq \inf_{f \in \mathcal{A}} \mathcal{E}(f),$$

Let $\lambda = \inf_{f \in \mathcal{A}} \mathcal{E}(f)$ and $\varepsilon > 0$. We recall that smooth functions on $[0, 1]$ are dense in $H_w^1(0, 1)$ (since it has a polynomial basis, see Appendix). Choose $f \in \mathcal{A} \cap C^\infty[0, 1]$ such that $\mathcal{E}(f) < \lambda + \varepsilon$. For each n , define $u_n \in \mathbb{R}^n$ by sampling f uniformly. That is,

$$u_n(k) := \frac{1}{n-1} f\left(\frac{k}{n-1}\right), \quad k = 0, \dots, n-1.$$

Then

$$\mathbf{1}^T u_n = \sum_{k=0}^{n-1} \frac{1}{n-1} f\left(\frac{k}{n-1}\right) \rightarrow \int_0^1 f(x) dx \quad (n \rightarrow \infty),$$

hence $|\mathbf{1}^T u_n| < \varepsilon$ holds for any sufficiently large n . Let us consider the sequence $u_n - \frac{1}{n} \mathbf{1}^T u_n$, which is clearly orthogonal to $\mathbf{1}$. Note that the weights $w_{k,k+1}$ are the midpoint values of the function $x(1-x)$ on $(k/(n-1), (k+1)/(n-1))$. By the Lagrange Mean Value Theorem and the convergence of the Riemann sums,

$$\begin{aligned} (u_n - \frac{1}{n} \mathbf{1}^T u_n)^T L_n (u_n - \frac{1}{n} \mathbf{1}^T u_n) &= \sum_{k=0}^{n-2} w_{k,k+1} (u_n(k) - u_n(k+1))^2 \\ &= \frac{1}{(n-1)^4} \sum_{k=0}^{n-2} w_{k,k+1} f'(\xi_k)^2 \\ &= \frac{1}{(n-1)^3} \sum_{k=0}^{n-2} \frac{1}{(n-1)} w(\xi_k) f'(\xi_k)^2 + o(1/n^3) \\ &= \frac{1}{(n-1)^3} \int_0^1 w(x) f'(x)^2 dx + o(1/n^3). \end{aligned}$$

Moreover,

$$\begin{aligned} (u_n - \frac{1}{n} \mathbf{1}^T u_n)^T (u_n - \frac{1}{n} \mathbf{1}^T u_n) &= \sum_{k=0}^{n-1} u_n(k)^2 - \frac{2}{n} |\mathbf{1}^T u_n|^2 + \frac{1}{n} |\mathbf{1}^T u_n|^2 \\ &= \frac{1}{n-1} \int_0^1 f^2(x) dx - \frac{1}{n} |\mathbf{1}^T u_n|^2 + o(1/n). \end{aligned}$$

Thus, for any sufficiently large n , we obtain that

$$\frac{(u_n - \frac{1}{n} \mathbf{1}^T u_n)^T L_n (u_n - \frac{1}{n} \mathbf{1}^T u_n)}{(u_n - \frac{1}{n} \mathbf{1}^T u_n)^T (u_n - \frac{1}{n} \mathbf{1}^T u_n)} (n-1)^2 \leq \mathcal{E}(f) + O(\varepsilon).$$

Since $\varepsilon > 0$ is arbitrary, we conclude that

$$\limsup_{n \rightarrow \infty} \left((n-1)^2 \min_{u \perp \mathbf{1}, u \neq 0} \frac{u^T L_n u}{u^T u} \right) \leq \inf_{f \in \mathcal{A}} \mathcal{E}(f).$$

In the next step, we prove the corresponding lim inf inequality.

Step 2. We have

$$\inf_{f \in \mathcal{A}} \mathcal{E}(f) \leq \liminf_{n \rightarrow \infty} \left((n-1)^2 \min_{u \perp \mathbf{1}, u \neq 0} \frac{u^T L_n u}{u^T u} \right).$$

For each n , let u_n be a minimizer of the discrete Rayleigh quotient, normalized by $\|u_n\|_2 = 1$. We embed the rescaled vector $\sqrt{n-1}u_n \in \mathbb{R}^n$ into the weighted Sobolev space $H_w^1(0,1)$ by linear interpolation. That is, define the function U_n piecewise

$$U_n(x) := (n-1)\sqrt{n-1}(u_n(k+1) - u_n(k)) \left(x - \frac{k}{n-1}\right) + \sqrt{n-1}u_n(k),$$

if $\frac{k}{n-1} \leq x \leq \frac{k+1}{n-1}$, $k = 0, 1, \dots, n-2$. We prove that the function sequence $\{U_n\}_n$ is bounded in $H_w^1(0,1)$. We may therefore apply a Rellich-Kondrachov-type compactness argument to find a cluster point of $\{U_n\}_n$ in $L^2(0,1)$. In fact, a direct calculation shows

$$\begin{aligned} \int_0^1 x(1-x)U_n'(x)^2 dx &= \sum_{k=0}^{n-2} \int_{k/(n-1)}^{(k+1)/(n-1)} (n-1)^3 (u_n(k) - u_n(k+1))^2 (x - x^2) dx \\ &= \sum_{k=0}^{n-2} ((k+1/2)(n-k-3/2) - 1/12) (u_n(k+1) - u_n(k))^2 \end{aligned}$$

Furthermore, from Step 1 we have that

$$\sup_n (n-1)^2 u_n^T L_n u_n = \sup_n \sum_{k=0}^{n-2} (k+1/2)(n-k-3/2) (u_n(k+1) - u_n(k))^2 < \infty$$

which clearly implies that

$$\sup_n \|\sqrt{x(1-x)}U_n'\|_2 < \infty.$$

A straightforward computation gives

$$\int_0^1 U_n^2(x) dx = \frac{1}{3} \sum_{k=0}^{n-2} u_n^2(k) + u_n^2(k+1) + u_n(k)u_n(k+1).$$

Hence, $\|U_n\|_2 \leq 1$ from the Cauchy-Schwarz inequality. Moreover, $a^2 + b^2 + ab \geq (a^2 + b^2)/2$ for any real a and b , so $\|U_n\|_2^2 \geq 1/6$. Due to the linear degeneracy of the weight w at the end-points of the unit interval, the classical Rellich-Kondrachov compactness theorem does not apply directly. Nevertheless, we can still use a compactness result in this case (see Appendix). We obtain that the sequence $U_n \in H_w^1(0,1)$ has a subsequence converging to $U \in L^2(0,1)$ in $L^2(0,1)$. This U is not the zero function because

$$\|U\|_2 \geq \liminf_{n \rightarrow \infty} \|U_n\|_2 \geq 1/\sqrt{6}.$$

We also have that

$$\int_0^1 U_n(x) dx = \frac{1}{2\sqrt{n-1}} \sum_{k=0}^{n-2} u_n(k) + u_n(k+1) = -\frac{1}{2\sqrt{n-1}} (u_n(0) + u_n(n-1)) \rightarrow 0;$$

hence, $0 \neq U \in \mathcal{A}$. Since the numerator is a convex functional, it is weakly lower semi-continuous; moreover the subsequence $U_n \rightarrow U$ strongly in $L^2(0,1)$. Therefore, the Rayleigh quotient $\mathcal{E}$ is weakly lower semi-continuous at $U \neq 0$.

This implies

$$\mathcal{E}(U) \leq \liminf_{n \rightarrow \infty} \mathcal{E}(U_n).$$

Thus, one can readily see that

$$\inf_{f \in \mathcal{A}} \mathcal{E}(f) \leq \liminf_{n \rightarrow \infty} \left((n-1)^2 \min_{u \perp 1, u \neq 0} \frac{u^T L_n u}{u^T u} \right).$$

Furthermore, from Step 1 we also obtain

$$\lim_{n \rightarrow \infty} \left((n-1)^2 \min_{u \perp 1, u \neq 0} \frac{u^T L_n u}{u^T u} \right) = \inf_{f \in \mathcal{A}} \mathcal{E}(f).$$

Note that the argument above, together with Step 1, shows in fact that the infimum is attained at U

$$\inf_{f \in \mathcal{A}} \mathcal{E}(f) = \min_{f \in \mathcal{A}} \mathcal{E}(f) = \mathcal{E}(U).$$

Let $\lambda = \mathcal{E}(U)$, and, from now on, let us use the notation $u_0 \equiv U$, for simplicity.

Step 3. Following the standard approach with the Euler-Lagrange differential equation, we solve the variational problem that leads us to a (singular) Sturm-Liouville differential equation. In fact, for any fixed $v \in \mathcal{A}$, $u_0 + \varepsilon v \in \mathcal{A}$. Furthermore, from Step 1 and Step 2, the function

$$E(\varepsilon) = \frac{\int_0^1 w(x)(u_0 + \varepsilon v)'(x)^2 dx}{\int_0^1 (u_0 + \varepsilon v)^2(x) dx}$$

attains its minimum value at $\varepsilon = 0$, and therefore $E'(0) = 0$. A direct calculation shows that

$$E'(0) = \frac{2 \int_0^1 w u'_0 v' dx \int_0^1 u_0^2 dx - 2 \int_0^1 w u_0'^2 dx \int_0^1 u_0 v dx}{\left(\int_0^1 u_0^2 dx \right)^2},$$

which vanishes if and only if

$$\int_0^1 w u'_0 v' dx = \frac{\int_0^1 w (u'_0)^2 dx}{\int_0^1 u_0^2 dx} \int_0^1 u_0 v dx = \mathcal{E}(u_0) \int_0^1 u_0 v dx.$$

Moreover, assuming v is smooth with compact support,

$$\int_0^1 w u'_0 v' dx = -[D(w u'_0), v],$$

in the distributional sense. Hence

$$[D(w u'_0) + \lambda u_0, v] = 0 \text{ for any } v \in C_c^\infty(0, 1) \text{ with } \int_0^1 v dx = 0.$$

This immediately implies that $D(wu'_0) + \lambda u_0 = c$ is the constant function. (Note that if $v \in C_c^\infty$ then $\int_0^1 v' dx = 0$.) In addition, since $u_0 \in H_w^1(0, 1)$ is absolutely continuous on any closed interval of $(0, 1)$ and $D(wu'_0) = c - \lambda u_0$, we also get that wu'_0 is continuously differentiable in $(0, 1)$. Since $1/w \in C^\infty(0, 1)$, it follows that $u_0 \in C^2(0, 1)$ must hold.

Let us now prove that $(wu'_0)' + \lambda u_0 = 0$ on $(0, 1)$; that is, $c = 0$. First, note that $wu'_0 \in H^1(0, 1)$. Indeed, $(wu'_0)' = c - \lambda u_0 \in L^2(0, 1)$, since $u_0 \in L^2(0, 1)$. Moreover, $wu'_0 \in L^2(0, 1)$ because $w^2(u'_0)^2 \leq w(u'_0)^2 \in L^2(0, 1)$. Therefore, $wu'_0 \in H^1(0, 1) \subseteq C[0, 1]$, which means that wu'_0 has a continuous extension to the closed interval $[0, 1]$; thus

$$0 = \int_0^1 wu'_0 v' dx - \lambda \int_0^1 u_0 v dx = wu'_0 v|_0^1 - \int_0^1 ((wu'_0)' + \lambda u_0) v dx = wu'_0 v|_0^1$$

for any $v \in \mathcal{A}$. As the endpoint values of $v \in \mathcal{A}$ can be arbitrary, we conclude that $wu'_0(0) = wu'_0(1) = 0$ must follow. Now

$$c = \int_0^1 (wu'_0)' + \lambda u_0 dx = \int_0^1 (wu'_0)' dx = wu'_0|_0^1 = 0.$$

Thus, u_0 is an eigenfunction (with eigenvalue λ) of the Sturm-Liouville problem

$$(x(1-x)u')' + \mu u = 0, \quad u \in C^2(0, 1) \cap \mathcal{A},$$

where μ is a real parameter.

We now determine the value of λ by transforming the Sturm-Liouville problem into the classical Legendre equation. This is the final step of the proof.

Step 4. Using the substitution, $t = 2x - 1$, we reduce the problem to Legendre's differential equation

$$(1-t^2)y'' - 2ty' + \mu y = 0;$$

that is,

$$((1-t^2)y')' + \mu y = 0,$$

where $y \in C^2(-1, 1) \cap L^2(-1, 1)$ and $\sqrt{1-t^2}y' \in L^2(-1, 1)$ and $\int_{-1}^1 y dt = 0$. We recall that the finite-energy solutions of Legendre equation are precisely the Legendre polynomials with $\mu = n(n+1)$, $n = 0, 1, 2, \dots$. This follows from the fact that the Legendre polynomials form an orthogonal basis of the weighted Sobolev space $H_{1-x^2}^1(-1, 1)$. For instance, the first few Legendre polynomials are

$$\mathcal{L}_0(t) = 1, \mathcal{L}_1(t) = t \text{ and } \mathcal{L}_2(t) = \frac{1}{2}(3t^2 - 1),$$

where $\mu = 0, 2$ and 6 , respectively. We claim that λ is first nonzero eigenvalue; i.e. $\lambda = 2$. Indeed, for any non-constant Legendre polynomial $\mathcal{L}_n$ with eigenvalue μ , we have

$$\lambda = \min_{f \in \mathcal{A}} \mathcal{E}(f) \leq \frac{\int_{-1}^1 (1-x^2)(\mathcal{L}'_n(x))^2 dx}{\int_{-1}^1 \mathcal{L}_n^2(x) dx} = \frac{\int_{-1}^1 ((1-x^2)\mathcal{L}'_n(x))' \mathcal{L}_n(x) dx}{\int_{-1}^1 \mathcal{L}_n^2(x) dx} = \mu.$$

Since λ is an eigenvalue and non-zero, we obtain that $\lambda = 2$, and

$$\lambda_2(L_n) \sim \frac{2}{(n-1)^2} \text{ as } n \rightarrow \infty.$$

Appendix

Rellich-Kondrachov-type compactness theorem in $H_w^1(0, 1)$. Let f_n be a bounded sequence in $H_w^1(0, 1)$. Then it has a convergent subsequence in $L^2(0, 1)$.

Proof. Let us define the function sequence f_n on the whole real line by 0 outside of the unit interval. We directly apply the Kolmogorov-Riesz theorem to prove precompactness in $L^2(0, 1)$. In practice, it means that it is enough to check that translation is uniformly continuous

$$\sup_n \int_0^1 (f_n(x+h) - f_n(x))^2 dx \rightarrow 0, h \rightarrow 0.$$

Clearly,

$$\sup_n \int_0^1 (f_n(x+h) - f_n(x))^2 dx \leq \sup_n \int_0^{1-h} (f_n(x+h) - f_n(x))^2 dx + \sup_n \int_{[0,1] \setminus [h,1-h]} f_n^2(x) dx$$

holds. For any locally absolutely continuous function f ,

$$\begin{aligned} f(x) - f(y) &= \int_y^x f'(s) ds = \int_y^x f'(s) \sqrt{w(s)} \frac{1}{\sqrt{w(s)}} ds \\ &\leq \left(\int_0^1 f'(s)^2 s(1-s) ds \right)^{1/2} \left(\int_y^x \frac{1}{w(s)} ds \right)^{1/2}. \end{aligned}$$

Hence,

$$\begin{aligned} \int_0^{1-h} (f_n(x+h) - f_n(x))^2 dx &\leq \|f_n\|_{H_w^1(0,1)}^2 \int_0^{1-h} \int_x^{x+h} \frac{1}{s(1-s)} ds dx \\ &= -2\|f_n\|_{H_w^1(0,1)}^2 (h \ln h + (1-h) \ln(1-h)) \rightarrow 0, h \rightarrow 0. \end{aligned}$$

Moreover, without loss of generality, we can assume that $\int_0^1 f_n dx = 0$ by decomposing the sequence $f_n = \left(f_n - \int_0^1 f_n dx\right) + \int_0^1 f_n dx$. In fact, by the Cauchy-Schwarz inequality, the scalar sequence $\int_0^1 f_n dx$ is bounded, hence it has a convergent subsequence. For any such $f = f_n$,

$$f^2(x) = \left(\int_0^1 f(x) - f(y) dy \right)^2 \leq \int_0^1 (f(x) - f(y))^2 dy$$

$$\leq \left(\int_0^1 f'(s)^2 s(1-s) ds \right) \int_0^1 \int_{\min(x,y)}^{\max(x,y)} \frac{1}{w(s)} ds dy;$$

For any fixed $0 < x < 1$, a direct calculation gives

$$\begin{aligned} \int_0^1 \int_{\min(x,y)}^{\max(x,y)} \frac{1}{w(s)} ds dy &= \int_0^x \int_y^x \frac{1}{s(1-s)} ds dy + \int_x^1 \int_x^y \frac{1}{s(1-s)} ds dy \\ &= -\log x(1-x). \end{aligned}$$

Thus

$$\begin{aligned} \int_{[0,h] \cup [1-h,1]} f_n^2(x) dx &\leq -\|f_n\|_{H_w^1(0,1)}^2 \int_{[0,h] \cup [1-h,1]} \log x(1-x) dx \\ &= \|f_n\|_{H_w^1(0,1)}^2 (4h + 2(1-h) \log 1 - h) - 2h \log h \rightarrow 0, h \rightarrow 0, \end{aligned}$$

which completes the proof.

The Legendre polynomials form an orthogonal basis of the weighted Sobolev space $H_{1-x^2}^1(-1, 1)$.

Proof. First, recall that the $\mathcal{L}_n$ Legendre polynomials form an orthogonal basis of $L^2(-1, 1)$. Indeed, it is well-known that the Gram-Schmidt orthogonalization process applied to the natural complete system $\{1, x, x^2, \dots\}$ of $L^2(-1, 1)$ defines the Legendre polynomials. Now, let us choose an $f \in H_{1-x^2}^1(-1, 1)$ such that

$$0 = (f, \mathcal{L}_n)_{H_{1-x^2}^1} = \int_{-1}^1 f(x) \mathcal{L}_n(x) dx + \int_{-1}^1 f'(x) \mathcal{L}_n'(x) (1-x^2) dx$$

holds for every $n \geq 0$. We claim that $f = 0$ a.e.. Note that

$$f(x)(1-x^2) \in W^{1,1}(-1, 1) \subseteq C[-1, 1].$$

Indeed, $(f(x)(1-x^2))' = f'(x)(1-x^2) - 2xf(x)$ in the weak sense, and the difference readily belongs to $L^2(-1, 1)$. Hence, $f(x)(1-x^2)$ has continuous extension to the closed interval $[-1, 1]$. An indirect argument shows that $f(x)(1-x^2)$ must be 0 at the end-points. (If $\lim_{x \rightarrow \pm 1} f(x)(1-x^2) = c_{\pm}$ then $f(x) \sim c_{\pm}/(1-x^2)$ in the vicinity of the end-points. This belongs to $L^2(-1, 1)$ if and only if $c_{\pm} = 0$.) Therefore, we can apply integration by parts and obtain

$$\int_{-1}^1 f'(x) \mathcal{L}_n'(x) (1-x^2) dx = - \int_{-1}^1 f(x) (\mathcal{L}_n'(x) (1-x^2))' dx = n(n+1) \int_{-1}^1 f(x) \mathcal{L}_n(x) dx,$$

because $\mathcal{L}_n$ satisfies the corresponding Legendre ODE. Thus

$$0 = (f, \mathcal{L}_n)_{H^1} = (1 + n(n+1)) \int_{-1}^1 f(x) \mathcal{L}_n(x) dx.$$

Since $f \in L^2(-1, 1)$ is orthogonal to each basis element $\mathcal{L}_n(x)$ of $L^2(-1, 1)$, we obtain that $f = 0$ a.e.. Moreover, a straightforward computation shows that $\mathcal{L}_n(x)$, $n = 0, 1, \dots$, are orthogonal in $H_{1-x^2}^1(-1, 1)$ which completes the proof. By a simple change of variables, we also obtain that $\mathcal{L}_n(x/2 + 1/2)$, $n = 0, 1, \dots$, form an orthogonal basis of $H_w^1(0, 1)$.